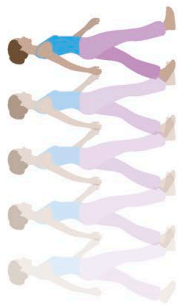


SLEEP DISORDERS

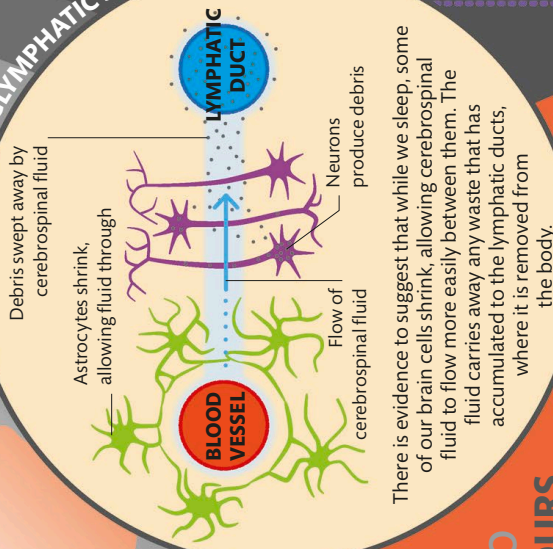
Problems like sleepwalking, sleep talking, and paralysis occur when the brain fails to make a clean shift between sleep states. This leaves part of our brain awake while other parts are sound asleep. When a person sleepwalks, the motor areas of the brain are awake and active, but the conscious awareness and memory areas are asleep. People can even perform complex tasks such as driving while fast asleep.



Cleaning the brain

During the day, our brain activity produces by-products that can become toxic if they build up. Recent studies using mice have shown that sleep gives the brain a chance to clean these by-products away. It seems likely that something similar happens in humans, which may explain some of the negative effects sleep deprivation can have on our ability to learn, remember, and manage our emotions.

THE GLYPHATIC SYSTEM



THE LONGEST RECORDED ATTEMPT TO STAY AWAKE IS 264 HOURS



The dreaming brain

Scientists do not know why we dream, but they have theories. Dreams might help us process information and emotions encountered during the day and store them in our long-term memory (see pp.138–139). A dream might also be like a rehearsal—our brain is trying out responses to extreme events in safety so we would be prepared if the event happened in real life. This might explain why dreams are often stressful or negative. Another idea is that dreams are merely “screen savers” for the mind, with no real purpose at all.

